

References

- Zied Bouraoui, Jose Camacho-Collados, and Steven Schockaert. 2019. Inducing relational knowledge from BERT. In *AAAI*.
- Samuel R Bowman, Gabor Angeli, Christopher Potts, and Christopher D Manning. 2015. A large annotated corpus for learning natural language inference. In *EMNLP*.
- Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*.
- Alexis Conneau, Germán Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. 2018. What you can cram into a single vector: Probing sentence embeddings for linguistic properties. In *ACL*.
- Ido Dagan, Oren Glickman, and Bernardo Magnini. 2005. The PASCAL recognising textual entailment challenge. In *Machine Learning Challenges Workshop*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: pre-training of deep bidirectional transformers for language understanding. In *NAACL*.
- Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. 2020. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*.
- Hady ElSahar, Pavlos Vougiouklis, Arslan Remaci, Christophe Gravier, Jonathon S. Hare, Frédérique Laforest, and Elena Simperl. 2018. T-REx: A large scale alignment of natural language with knowledge base triples. In *LREC*.
- John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. In *EMNLP*.
- Shankar Iyer, Nikhil Dandekar, and Kornel Csernai. 2017. [First quora dataset release: Question pairs](#).
- Sarthak Jain and Byron C Wallace. 2019. Attention is not explanation. In *NAACL*.
- Zhengbao Jiang, Frank F Xu, Jun Araki, and Graham Neubig. 2020. How can we know what language models know? In *TACL*.
- Sunjae Kwon, Cheongwoong Kang, Jiyeon Han, and Jaesik Choi. 2019. Why do masked neural language models still need common sense knowledge? *arXiv preprint arXiv:1911.03024*.
- Patrick Lewis, Ludovic Denoyer, and Sebastian Riedel. 2019. Unsupervised question answering by cloze translation. In *ACL*.
- Nelson F Liu, Matt Gardner, Yonatan Belinkov, Matthew Peters, and Noah A Smith. 2019. Linguistic knowledge and transferability of contextual representations. In *NAACL*.
- Ilya Loshchilov and Frank Hutter. 2018. Decoupled weight decay regularization. In *International Conference on Learning Representations*.
- Marco Marelli, Stefano Menini, Marco Baroni, Luisa Bentivogli, Raffaella Bernardi, Roberto Zamparelli, et al. 2014. A SICK cure for the evaluation of compositional distributional semantic models. In *LREC*.
- Marius Mosbach, Maksym Andriushchenko, and Dietrich Klakow. 2020. [On the stability of fine-tuning bert: Misconceptions, explanations, and strong baselines](#).
- Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. In *NAACL*.
- Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H Miller, and Sebastian Riedel. 2019. Language models as knowledge bases? In *EMNLP*.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *Technical report*.
- Timo Schick and Hinrich Schütze. 2020. Exploiting cloze questions for few-shot text classification and natural language inference. *arXiv preprint arXiv:2001.07676*.
- Vered Shwartz, Peter West, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2020. Unsupervised commonsense question answering with self-talk. *arXiv preprint arXiv:2004.05483*.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *EMNLP*.
- Daniil Sorokin and Iryna Gurevych. 2017. Context-aware representations for knowledge base relation extraction. In *EMNLP*.
- Trieu H Trinh and Quoc V Le. 2018. A simple method for commonsense reasoning. *arXiv preprint arXiv:1806.02847*.
- Elena Voita and Ivan Titov. 2020. Information-theoretic probing with minimum description length. In *EMNLP*.
- Eric Wallace, Shi Feng, Nikhil Kandpal, Matt Gardner, and Sameer Singh. 2019. Universal adversarial triggers for attacking and analyzing NLP. In *EMNLP*.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R Bowman. 2019. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *ICLR*.

Sarah Wiegreffe and Yuval Pinter. 2019. Attention is not not explanation. In *EMNLP*.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. 2019. HuggingFace’s Transformers: State-of-the-art natural language processing. *arXiv preprint arXiv:1910.03771*.

A Effect of Hyperparameters on Sentiment Analysis

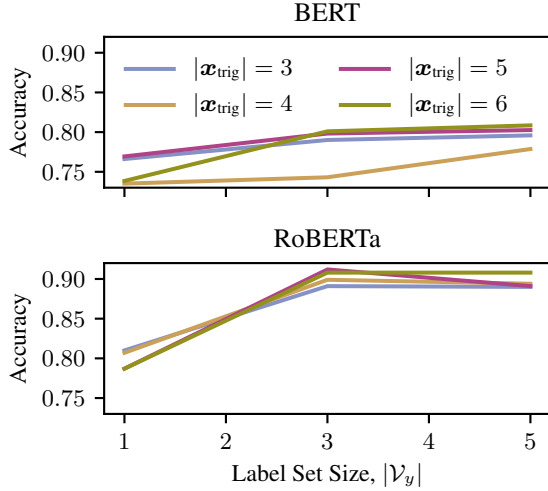


Figure 3: **Effect of Label and Trigger Set Sizes** on sentiment analysis. The number of candidate replacements is fixed at $|\mathcal{V}_{\text{cand}}| = 100$. Increasing the label set size improves performance, while changing the trigger length does not have much impact.

To measure the effects of the AUTOPROMPT search hyperparameters, we plot the validation accuracy as a function of label set size $|\mathcal{V}_y|$ and the number of trigger tokens $|x_{\text{trig}}|$ in Figure 3. We fix the number of candidates at $|\mathcal{V}_{\text{cand}}| = 100$. We observe similar trends when $|\mathcal{V}_{\text{cand}}| = 10$.

Varying the number of trigger tokens generally has little effect. On the other hand, there is a substantial increase in accuracy when increasing the label set size from 1 to 3 (approximately +5% for BERT, and +10% for RoBERTa). After analyzing the label sets, we find that our method generally produces intuitive results—“marvelous” and “philanthrop” are associated with positive sentiment, whereas “worse” and “incompetence” are associated with negative sentiment for RoBERTa.

B Relation Extraction Details

Following [Petroni et al. \(2019\)](#), we use the pre-trained RE model from [Sorokin and Gurevych \(2017\)](#) as our baseline. To encode the sentence, this model uses a combination of an LSTM-based relation encoder and an attention mechanism. To make predictions, the model constructs a knowledge graph whose edges are the extracted relation triples. The standard RE evaluation measures how well the model predicts the relation types of entity pairs on the sentence level.

Since our goal is to extract the object of relation triplets, rather than the relation itself, we tweak the standard RE evaluation. We feed the RE model sentences from test facts and we query the resulting graph for all edges that contain the given subject and relation. Then we select the triple with the highest confidence and compare it’s object to the gold object. We do this for every fact and take the average across all relations to get the overall precision. The RE model is not trained to predict two of the original T-REx relations. For fair comparison, we exclude these two relations for our evaluation.